Topology-Preserving Contrastive Learning for Taxonomy Induction in Thai Lexical Semantics

Thai WordNet

Escaping the Symmetric Trap in Lexical Semantics

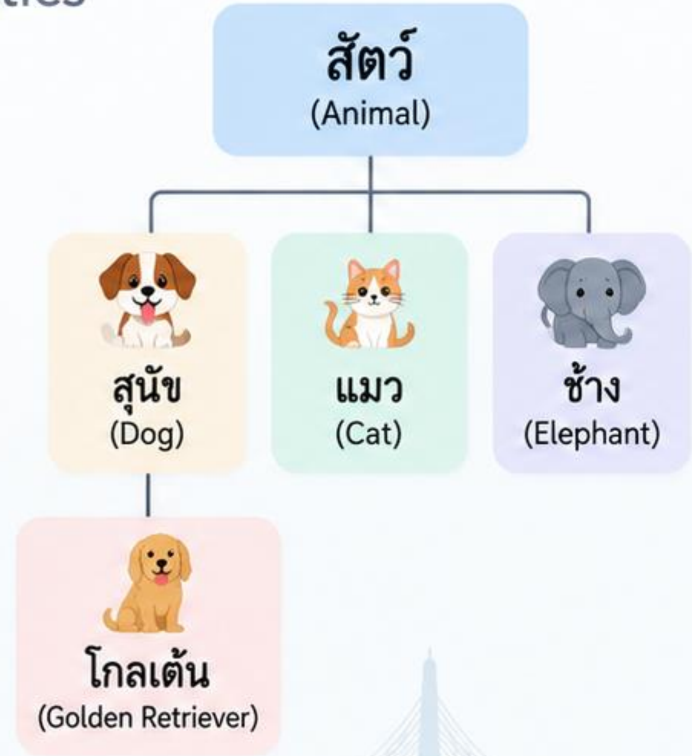
Chayanapas Kaoien[1], Dhanon Leenoi[2], Peerachet Porkaew[2], and Vorapon Luantangsrisuk[2]

[1] Faculty of Engineering and Technology, Panyapiwat Institute of Management, Nonthaburi, Thailand

[2] National Electronics and Computer Technology Center (NECTEC), Pathum Thani, Thailand

PANYAPIWAT INSTITUTE OF MANAGEMENT

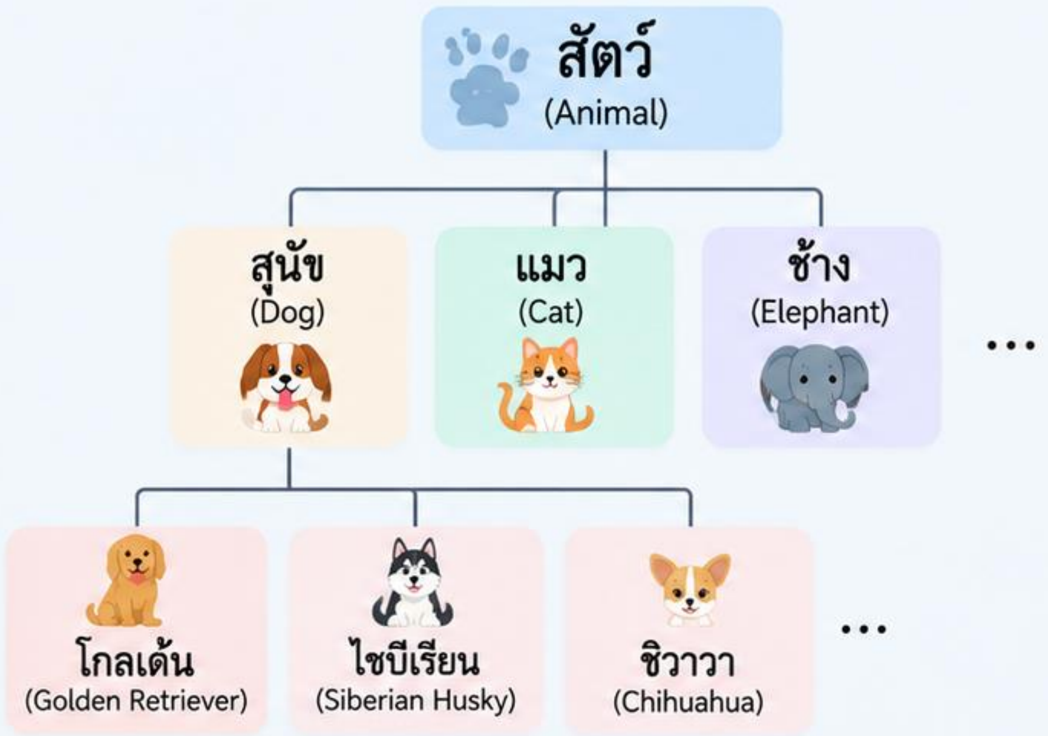
NECTEC a member of NSTDA



Learning that two concepts are **related** is not the same as learning their **hierarchy**.

Motivation: Why Taxonomy?

A taxonomy organizes concepts into a structured, hierarchical map of knowledge.



Structured Knowledge
Taxonomy organizes concepts from general to specific.

Supports Many Applications
Useful for semantic search, conversational AI, and retrieval-augmented generation (RAG).

Costly to Build Manually
Manual curation is time-consuming and expensive.

Needs Automatic Induction
We need methods that can automatically learn lexical relations from resources such as Thai WordNet.

> *A Knowledge Graph is not just a collection of similar words. It is a **structured map of concepts**.*

Real-world Applications

Semantic Search
เช่น ค้นหาข้อมูลที่เกี่ยวข้อง

Conversational Agents
เช่น ตอบคำถามอย่างมีบริบท

Retrieval-Augmented Generation (RAG)
เช่น ดึงความรู้จากกราฟความรู้

Downstream NLP Tasks
เช่น การจัดหมวดหมู่, การแปล

Taxonomy provides the backbone for **intelligent language understanding**.

Thai Example: สัตว์ → สุนัข

A simple example shows why taxonomy is hierarchical and directional.

Taxonomy Hierarchy (Thai WordNet)

- สัตว์ (Animal)
  - สุนัข (Dog)
    - โกลเด้น (Golden Retriever)
    - ไซบีเรียน (Siberian Husky)
    - ชิวาวา (Chihuahua)
    - ...
  - แมว (Cat)
  - ช้าง (Elephant)
  - ...

IS-A Relation (Hypernym)

สัตว์ (Animal) → **สุนัข** (Dog)

สัตว์ เป็นคำที่มีความหมายกว้างกว่า
และ **สุนัข** เป็นคำที่มีความหมายเฉพาะกว่า
“สุนัข เป็นสัตว์ชนิดหนึ่ง”

Inverse Relation (Hyponym)

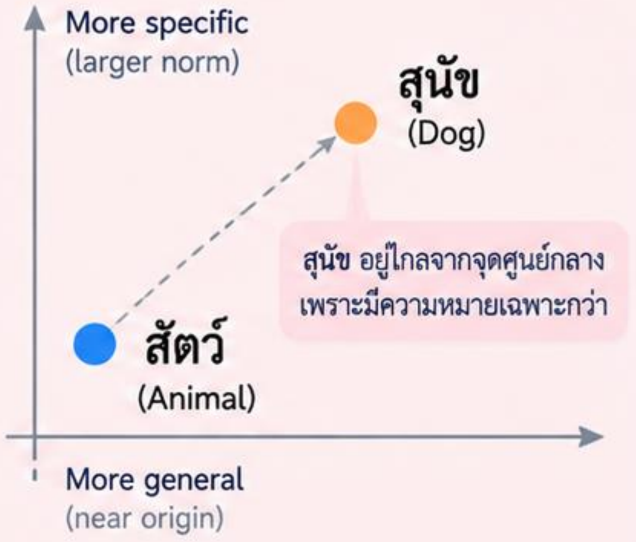
สุนัข (Dog) → **สัตว์** (Animal)

สุนัข เป็นชนิดย่อย (hyponym) ของ **สัตว์**
“สุนัข จัดเป็นสัตว์”

Key Point

ทั้งสองคำ “เกี่ยวข้องกัน” แต่มีทิศทาง ไม่ใช่สมมาตร
They are related, but the relation is **directional**.

Intuition in Embedding Space



แนวคิดของ LCCL:
คำที่มีความหมายทั่วไปอยู่ใกล้จุดศูนย์กลาง
และคำที่เฉพาะเจาะจงกว่าจะอยู่ไกลออกไป

“โครงสร้างแบบลำดับชั้น ช่วยให้เรามองเห็นความสัมพันธ์ของคำในภาพรวม”
Taxonomy helps us see the big picture of how concepts are related.

Symmetric vs. Asymmetric Relations

Not all lexical relations are the same. Some are symmetric, while others are directional.

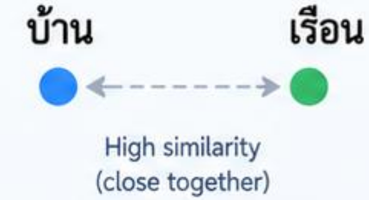
Symmetric Relations

Both directions hold (or the relation is inherently symmetric).

Synonym (คำพ้องความหมาย) บ้าน (house) ⟷ เรือน (house)

Antonym (คำตรงข้าม) ร้อน (hot) ⟷ เย็น (cold)

Embedding Space (Symmetric)



For **symmetric** relations, we can **pull** the embeddings **close together**.

≠

Are they close?

≠

Which one is the parent?

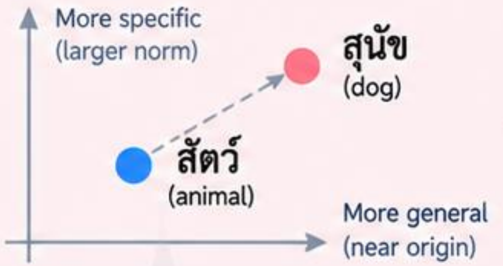
Asymmetric Relations (IS-A)

Only one direction holds. The relation is directional.

Hypernym (คำที่มีความหมายกว้างกว่า) สัตว์ (animal) → สุนัข (dog)

Hyponym (คำที่มีความหมายเฉพาะกว่า) สุนัข (dog) → สัตว์ (animal)

Embedding Space (Asymmetric)



For **IS-A relations, closeness is not enough.** We need to encode **direction** (e.g., parent → child) in the embedding space.

Semantic similarity tells us that words are **related**.
But taxonomy requires more — we must also capture the **direction** of the relation.

The Symmetric Trap

A symmetric similarity objective pulls directional relations into the same neighborhood, making the model know "they are close" but not "which one is the parent."

What a symmetric contrastive loss sees

Cosine Similarity

สัตว์
(Animal)

สุนัข
(Dog)

Very similar!

But the real relation is directional

สัตว์
(Parent / Hypernym)

↓

สุนัข
(Child / Hyponym)

The model learns "close", but not "which way."

Symmetric objective:
similarity(สัตว์, สุนัข) = similarity(สุนัข, สัตว์)
→ Directional information is lost.

More examples

Symmetric (Synonym)

บ้าน (house) ↔ เรือน (dwelling)

Close (both directions)

Symmetric (Antonym)

ร้อน (hot) ↔ เย็น (cold)

Far (push apart)

Asymmetric (Hypernym)

สัตว์ (animal) → สุนัข (dog)

Parent → Child
(not equivalent)

Asymmetric (Hyponym)

สุนัข (dog) → สัตว์ (animal)

Child → Parent
(not equivalent)

Symmetric similarity is not enough.
We need a way to represent direction and hierarchy in the embedding space.

ใกล้กัน... แต่ไม่เท่ากัน

Why Classification Accuracy Is **Not Enough**

A model can classify lexical relations correctly, but still fail to preserve the hierarchical structure in the embedding space.

High Classification Performance

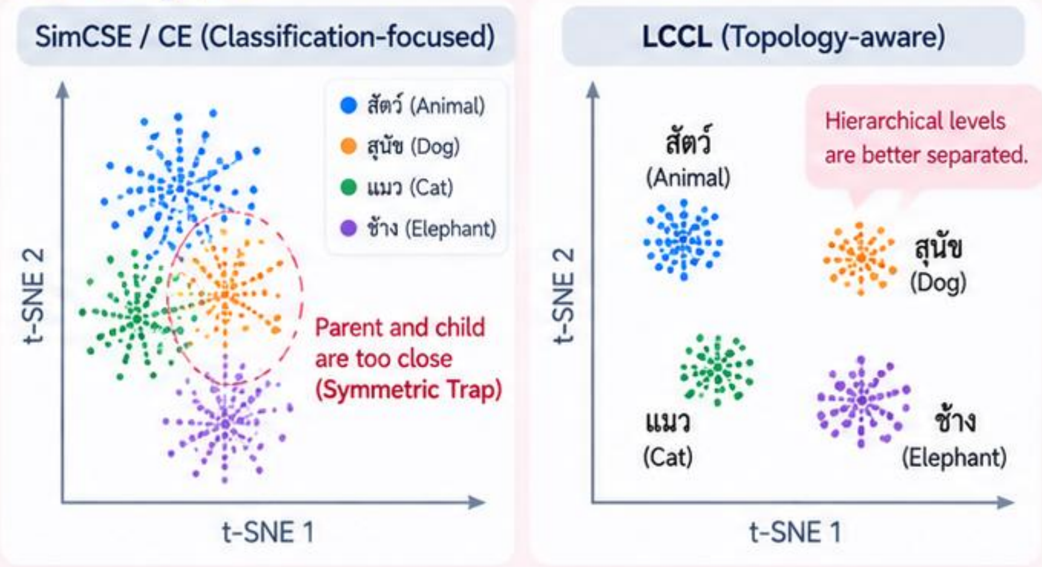
LCCL achieves comparable classification accuracy with strong baselines.

Model	Macro F1 (↑)
BERT (pairwise)	0.642
SimCSE + CE	0.667
Cross-Encoder	**0.681**
LCCL (Ours)	**0.676**

Classification accuracy alone suggests that the model understands lexical relations.

But the Embedding Geometry Can Collapse

Even with high accuracy, symmetric objectives may place parent and child concepts too close in the embedding space.



Impact on Downstream Tasks

When the hierarchy collapses, graph traversal and KG construction suffer, even if classification looks good.

สัตว์ (Animal)
↓
สุนัข (Dog)
↓
โกลเด้น (Golden Retriever)

If parent and child are indistinguishable in the embedding space, it is hard to:

- find the correct parent
- find the correct children
- recover missing edges
- build a reliable knowledge graph

Classification tells us **what relation a pair has**,
but topology preservation ensures the embedding space **keeps the hierarchy**.

LCCL: Core Idea

Learn lexical representations with selective objectives — similarity for symmetric relations, directional constraints for asymmetric relations.

1. Different Objectives for Different Relations

We apply relation-aware learning signals.

Symmetric relations

Use contrastive similarity (cosine loss).

Pull similar pairs together

บ้าน (house) ⟷ เรือน (house)

- Synonym (คำพ้องความหมาย)
- Antonym (คำตรงข้าม)
- Related (คำที่เกี่ยวข้อง)

Asymmetric relations (IS-A)

Use directional norm constraint ($\mathcal{L}_{\text{asym}}$).

Encode hierarchical direction

สัตว์ (animal) ⟶ สุนัข (dog)

- Hypernym (คำที่มีความหมายกว้างกว่า)
- Hyponym (คำที่มีความหมายเฉพาะกว่า)

2. Joint Learning Objective

LCCL combines classification and geometric objectives.

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{cos}}\mathcal{L}_{\text{cos}} + \lambda_{\text{asym}}\mathcal{L}_{\text{asym}}$$

- $\mathcal{L}_{\text{CE}}$: Classification Loss (multi-class)
- $\lambda_{\text{cos}}\mathcal{L}_{\text{cos}}$: Contrastive Cosine Loss (for symmetric & negative pairs)
- $\lambda_{\text{asym}}\mathcal{L}_{\text{asym}}$: Directional Norm Loss (for IS-A pairs)

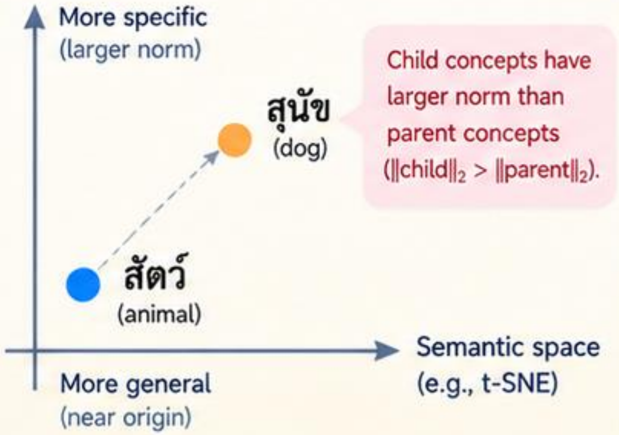
Selective Application (Masking)

Apply each loss only where it is topologically appropriate.

Relation Type	Classification ($\mathcal{L}_{\text{CE}}$)	Cosine Loss ($\mathcal{L}_{\text{cos}}$)	Asymmetric Loss ($\mathcal{L}_{\text{asym}}$)
Synonym	✓	✓	–
Antonym	✓	✓	–
Related	✓	✓	–
Hypernym (IS-A)	✓	–	✓
Hyponym (IS-A)	✓	–	✓
Negative (unrelated)	✓	✓	–

3. Desired Embedding Geometry

General concepts near the origin, specific concepts further away.



Key Intuition

Not just close, but ordered — embeddings reflect the hierarchy.

LCCL = Classification + Selective Geometry

Same encoder, two complementary learning signals — semantic accuracy and topological structure.

Two Geometric Objectives in LCCL

Use contrastive cosine loss for symmetric/negative relations, and an asymmetric norm constraint for IS-A relations.

1 Contrastive Cosine Loss

Used for symmetric relations (e.g., synonym, antonym, related) and negative pairs.

Pull similar pairs together, push dissimilar pairs apart.

$$\mathcal{L}_{\cos} = -y \cdot \log \sigma(\cos(u,v)) - (1-y) \cdot \log \sigma(-\cos(u,v))$$

u, v : embeddings of the two words
y : 1 for similar pairs, 0 for negative pairs

Similar pair (pull together)

บ้าน (house) ← → เรือน (house)

cosine ↑

Make them close

Dissimilar pair (push apart)

บ้าน (house) ← → รถยนต์ (car)

cosine ↓

Make them far apart

Capture semantic similarity for symmetric relations and keep unrelated pairs separated.

2 Asymmetric Norm Constraint

Used only for IS-A relations (hypernym → hyponym).

Enforce that child concepts have larger norm than their parent concepts.

$$\mathcal{L}_{\text{asym}} = \max(0, m + ||u||_2 - ||v||_2)$$

u : embedding of parent (hypernym)
v : embedding of child (hyponym)
m : margin (e.g., 0.1)

Hierarchical Structure in Embedding Space

Example: สัตว์ → สุนัข

$||\text{สัตว์}||_2 = 2.0$
$||\text{สุนัข}||_2 = 3.0$

$||\text{สุนัข}||_2 > ||\text{สัตว์}||_2$

Child is farther from the origin than parent.

Apply this constraint only to IS-A pairs. Do not use it for synonyms or other symmetric relations.

Contrastive cosine learns **what is close.**
Asymmetric norm constraint learns **which one is the parent.**

LCCL Architecture

A multilingual-e5-base encoder with multi-objective training to learn topology-aware lexical representations.

1 Input: Training Triplets

Create training examples from the taxonomy and a general corpus.

Symmetric pairs (similar)

บ้าน (house) ↔ เรือน (house) (synonym)

Negative pairs (dissimilar)

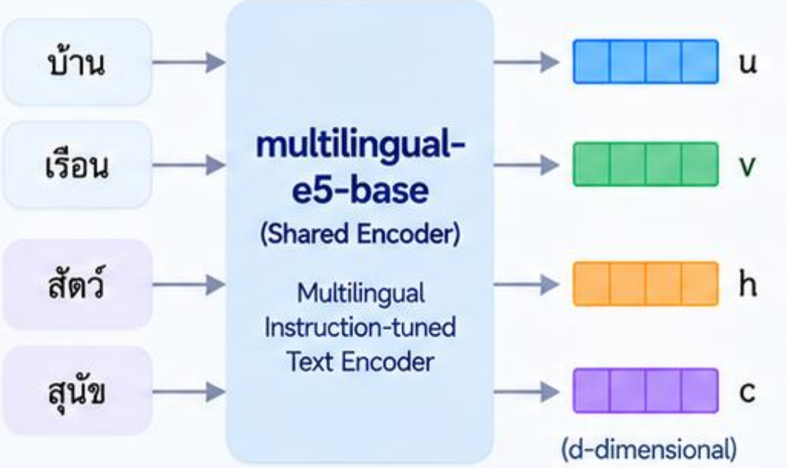
บ้าน (house) ↔ รถยนต์ (car) (unrelated)

IS-A pairs (asymmetric)

สัตว์ (animal) → สุนัข (dog) (hypernym → hyponym)

2 Model: Multilingual Encoder

Use **multilingual-e5-base** as the shared encoder to obtain sentence embeddings.



3 Training Objectives

Jointly optimize multiple losses.

Classification Loss $\mathcal{L}_{\text{CE}}$

Masked language modeling (next token prediction) on a general corpus.

Contrastive Cosine Loss $\mathcal{L}_{\text{cos}}$

Pull similar pairs together, push dissimilar pairs apart (symmetric relations).

$$\mathcal{L}_{\text{cos}} = -y \cdot \log \sigma(\cos(u, v)) - (1 - y) \cdot \log \sigma(-\cos(u, v))$$

Asymmetric Norm Loss $\mathcal{L}_{\text{asym}}$

Enforce that child concepts have larger norm than parents (IS-A relations).

$$\mathcal{L}_{\text{asym}} = \max(0, m + \|u\|_2 - \|v\|_2)$$

4 Overall Training Objective

Combine all losses with weighting coefficients.

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{cos}}\mathcal{L}_{\text{cos}} + \lambda_{\text{asym}}\mathcal{L}_{\text{asym}}$$

$\lambda_{\text{cos}}, \lambda_{\text{asym}}$ control the contribution of each objective.

A single **multilingual-e5-base** encoder learns both **semantic accuracy** and **topology-preserving structure**.

LCCL: Training Overview

Jointly optimize classification and geometric objectives to learn topology-aware lexical representations.

1 Input: Word Pair (u, v)

Sample training pairs from the taxonomy (e.g., synonym, antonym, IS-A, unrelated).

2 Multilingual Encoder

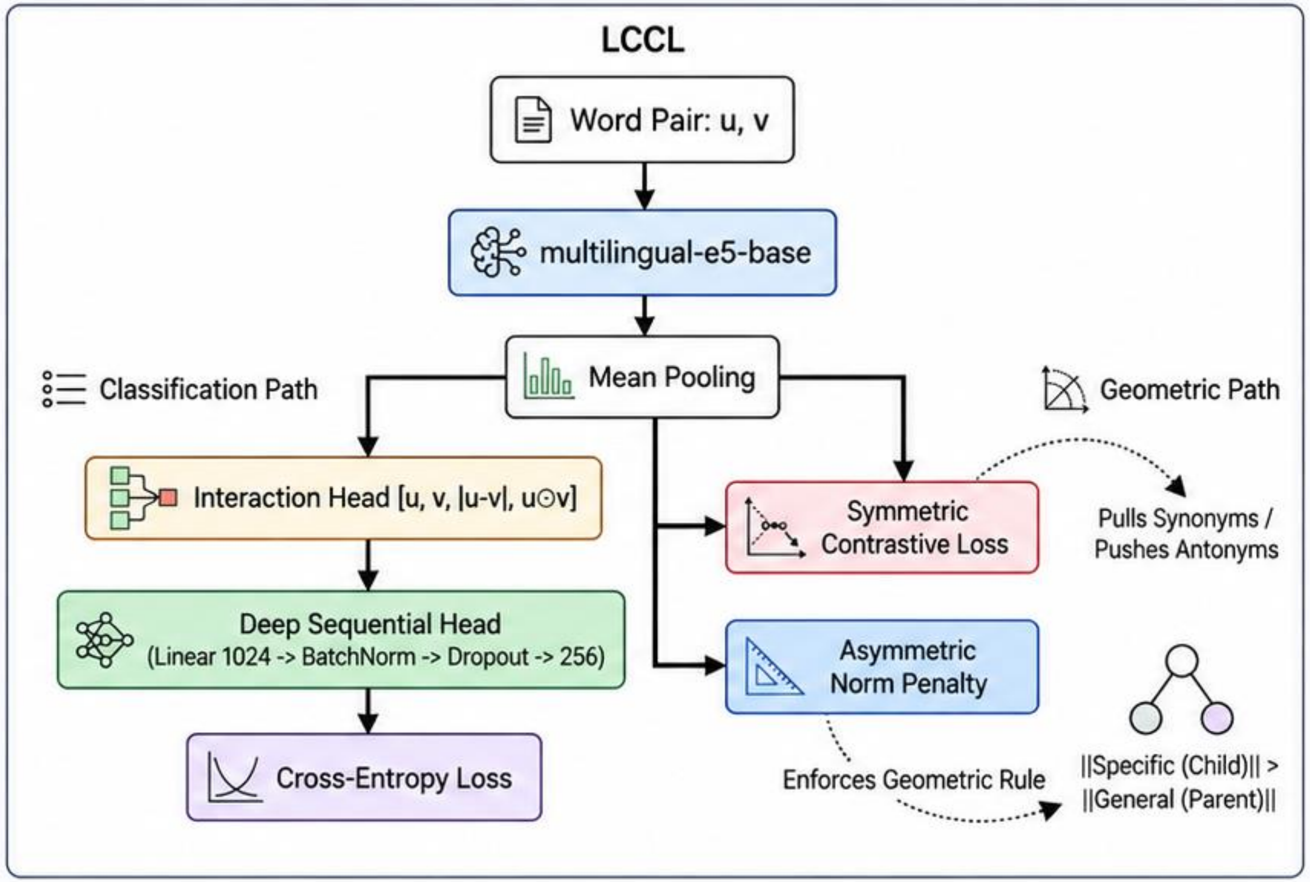
Use **multilingual-e5-base** to encode each word into dense embeddings.

3 Mean Pooling

Apply mean pooling to obtain fixed-size vector representations for u and v.

4 Classification Path

Use an interaction head and a deep sequential head to predict the relation type with cross-entropy loss.



5 Geometric Objectives

Use contrastive cosine loss for symmetric relations and an asymmetric norm penalty for IS-A relations.

6 Joint Training

Optimize all objectives together with weighting coefficients.

$$\mathcal{L}_{Total} = \mathcal{L}_{CE} + \lambda_{cos}\mathcal{L}_{cos} + \lambda_{asym}\mathcal{L}_{asym}$$

- $\mathcal{L}_{CE}$: Classification loss (relation type)
- $\mathcal{L}_{cos}$: Contrastive cosine loss (symmetric relations)
- $\mathcal{L}_{asym}$: Asymmetric norm penalty (IS-A)
- $\lambda_{cos}, \lambda_{asym}$: Loss weights

One model, two complementary signals — learn **lexical meaning** and preserve the **taxonomy structure**.

Experimental **Setup**

Dataset statistics, training settings, and evaluation metrics.

1 Dataset Statistics

Thai lexical relations with train/validation splits and topology types.

Relation	Train	Val	Topology	Example (Thai)
Synonym	2,184	546	Symmetric	บ้าน - เรือน (House)
Antonym	556	139	Symmetric	ร้อน - เย็น (Hot-Cold)
Hypernym	5,848	1,462	Asymmetric	สัตว์ - สุนัข (Animal-Dog)
Hyponym	5,848	1,462	Asymmetric	สุนัข - สัตว์ (Dog-Animal)
Holonym	220	55	Asymmetric	รถ - ล้อ (Car-Wheel)
Meronym	224	56	Asymmetric	ล้อ - รถ (Wheel-Car)
Random	4,000	1,000	Disjoint	แมว - เก้าอี้ (Cat-Chair)
Total	**18,880**	**4,720**	-	-

2 Topology Types

Relations are grouped by their topological properties.

Symmetric
Interchangeable meanings
↔
e.g., synonym, antonym

Asymmetric
Directional (hypernym → hyponym)
→
e.g., hypernym, hyponym, holonym, meronym

Disjoint
Unrelated pairs
✕
e.g., random (unrelated)

3 Training Settings

Use multilingual-e5-base with the same settings for all methods.

Encoder	multilingual-e5-base (768-dim, mean pooling)
Input	Thai word/term (single token/phrase)
Max length	32 tokens
Batch size	256
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)

Learning rate	2×10^{-5} (with linear decay)
Training epochs	10
Loss (LCCL)	$\mathcal{L}_{total} = \mathcal{L}_{CE} + \lambda_{cos}\mathcal{L}_{cos} + \lambda_{asym}\mathcal{L}_{asym}$
$\lambda_{cos}, \lambda_{asym}$	0.5, 0.5 (tuned on dev set)
Hardware	1 × A100 40GB
Implementation	PyTorch + Hugging Face Transformers

4 Evaluation Metrics

We report standard metrics for relation classification and embedding quality.

Accuracy
Overall classification accuracy

Macro F1
Unweighted F1 across relation types

MRR
Mean reciprocal rank (for retrieval)

A well-balanced dataset with diverse lexical relations for fair comparison.

Experimental Results

LCCL produces better-separated embeddings than a standard baseline.

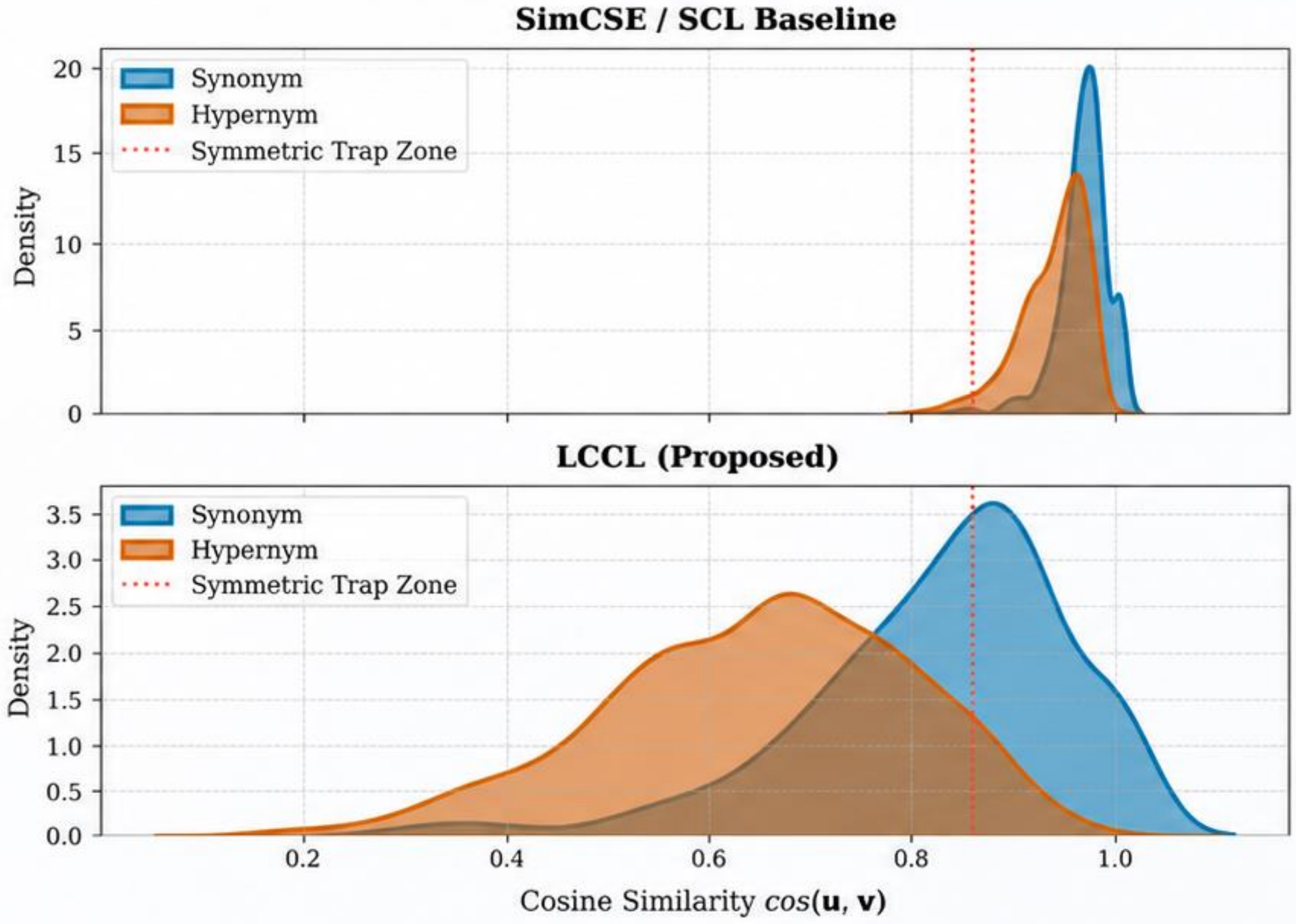


Fig. 2. Cosine similarity distributions (Synonym: blue; Hypernym: orange). SimCSE (top) shows severe overlap (KS = 0.412); LCCL (bottom) achieves clear separation (KS = 0.496).

Embedding Distributions

LCCL helps separate synonym and hypernym pairs in embedding space, reducing the overlap around the symmetric trap zone.

Key Observations

- SimCSE / SCL (top) shows severe everlap between synonym and hypernym distributions (KS = 0.412).
- LCCL (bottom) achieves clear separation (KS = 0.496).
- Synonym pairs are pushed to higher cosine similarity, while hypernym pairs are more spread and shifted to lower similarity.
- The symmetric trap zone (around 0.85) is effectively mitigated by LCCL.

Takeaway

By incorporating contrastive cosine loss and asymmetric norm constraint, LCCL learns embeddings that better reflect the lexical topology, leading to more distinguishable semantic relations.

Ablation Study

Effect of model components on Thai WordNet (23,600 pairs).

Table 2. Ablation Study on Thai WordNet (23,600 pairs). Semantic CE ignores graph topology. KS measures distributional separation between Synonym and Hypernym cosine similarities; larger values indicate stronger separation (used here as a geometric diagnostic). All baselines and LCCL are reported across five seeds with ± standard deviation to account for stochasticity. **Bold** = best per column.

Model	Loss	Acc.	Mac. F1	Hyper	Hypo	Holo	KS
Roller (2014)	CE	67.1%	0.599 ± 0.009	0.69	0.68	0.38	0.201
CE-only (Ours)	CE	70.4%	0.631 ± 0.011	0.66	0.70	0.42	0.287
Semantic CE	CE	**74.9%**	$\mathbf{0.681 \pm 0.006}$	**0.73**	**0.74**	0.49	0.318
SimCSE	$CE + \mathcal{L}_{cos}$	72.9%	0.667 ± 0.008	0.71	0.73	0.51	0.412
LEAR (2018)	CE + Norm	16.5%	0.040 ± 0.002	0.00	0.00	0.00	0.061
Poincaré	CE + Hyp.	16.5%	0.040 ± 0.002	0.00	0.00	0.00	0.006
LCCL (Ours)	$CE + \mathcal{L}_{cos,asym}$	72.9%	0.676 ± 0.008	0.72	0.73	**0.54**	**0.496**

LCCL Improves Separation

- LCCL achieves the highest KS (0.496), indicating the clearest separation between synonym and hypernym similarities.
- The improvement comes from incorporating contrastive cosine loss with asymmetric norm constraint.

Competitive Classification Performance

- Semantic CE yields the highest Acc. (74.9%) and Macro F1 (0.681), showing strong classification performance.
- LCCL achieves comparable Acc. (72.9%) and Macro F1 (0.676) while significantly improving topology-preserving separation (KS).

Takeaway

Combining contrastive cosine loss and asymmetric norm constraint helps LCCL learn embeddings that better reflect lexical topology, leading to more distinguishable semantic relations.

Conclusion / Takeaway

Topology-preserving contrastive learning for better lexical semantic embeddings.

Key Findings

LCCL effectively learns lexical embeddings that respect semantic topology.

1 Better separation of lexical relations

LCCL produces clearer separation between synonyms and hypernyms, as shown by higher KS (0.496 vs. 0.412 for SimCSE).

2 Competitive classification performance

LCCL achieves comparable accuracy (72.9%) and Macro F1 (0.676) to strong baselines, while significantly improving topology-preserving separation.

3 Effectiveness of proposed components

The combination of contrastive cosine loss and asymmetric norm constraint helps shape the embedding space according to lexical topology.

4 Thai WordNet resource

Our dataset of 23,600 word pairs across diverse lexical relations provides a useful benchmark for Thai lexical semantic representation learning.

Takeaway

What this work shows and why it matters.

- **Incorporating lexical topology into contrastive learning** leads to more structured and meaningful embeddings.
- **LCCL improves semantic separation** without sacrificing general language understanding.
- **The approach is simple, effective, and applicable** to real-world Thai NLP tasks such as semantic search, lexical substitution, and language understanding.

Future Work

Directions for further improvement.

- Extend to larger and more diverse lexical resources.
- Incorporate additional relational signals (e.g., meronymy, entailment).
- Apply to downstream tasks such as semantic textual similarity, lexical substitution, and information retrieval.

LCCL shows that modeling lexical topology in embedding spaces is a promising direction for richer and more interpretable lexical semantic representations in Thai.

Applications

Topology-aware embeddings enable practical Thai NLP applications.

1 Reversed Dictionary Search

Find words by meaning (not just by surface form).

"A word for a place where people live."

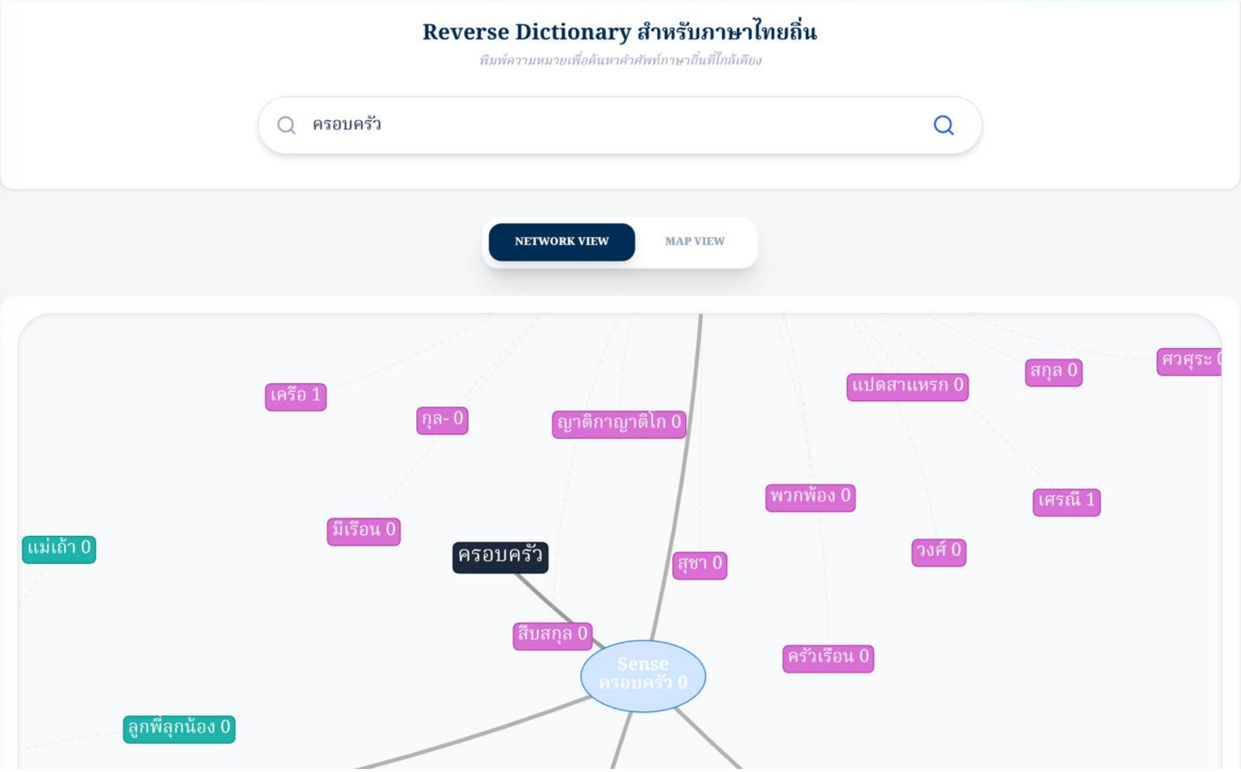
สถานที่ที่คนอาศัยอยู่ | Search

Top similar words (by meaning)

	Word	Meaning	Similarity
1	บ้าน	(house, home)	0.892
2	เรือน	(house, traditional Thai house)	0.851
3	ที่อยู่อาศัย	(residence)	0.823
4	คฤหาสน์	(mansion)	0.781
5	หอพัก	(dormitory)	0.755

2 Thai Dialects Word Search

Find words across Thai dialects and regional variations.



https://thdialects.pasaflow.com/

Thank you